

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	11 September 2026
Team ID	NA
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

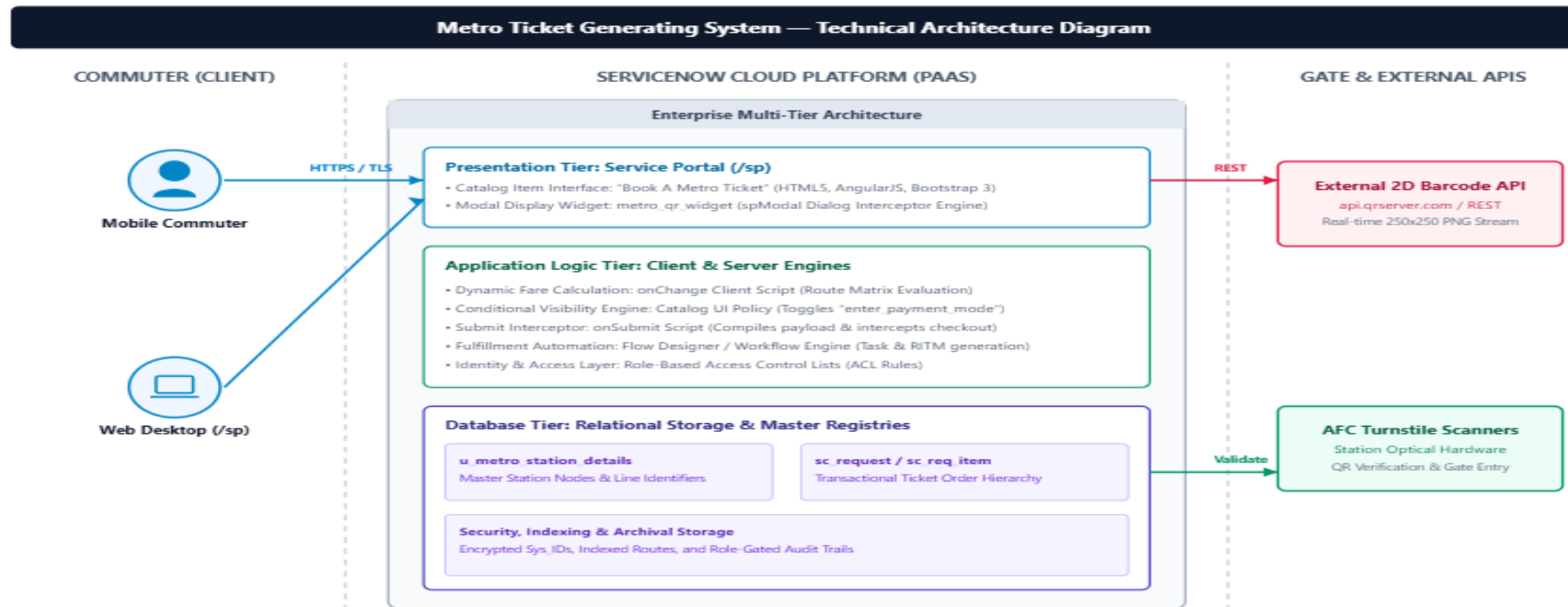


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web and mobile portal interface where commuters select stations, choose passenger counts, view dynamic fares, and inspect the generated QR pass.	HTML5, CSS3, JavaScript, AngularJS, Bootstrap 3 (ServiceNow Service Portal framework)
2.	Application Logic-1	Client-side dynamic fare computation matrix calculating single and return journey charges in real time upon route change.	JavaScript (ES5 / ES6), ServiceNow Catalog Client Script (onChange)
3.	Application Logic-2	Conditional UI Policy engine that shows and mandates alternate payment fields when "Others" is selected.	ServiceNow Catalog UI Policy & UI Policy Actions
4.	Application Logic-3	Submit interception engine that constructs the encoded payload, triggers ticket generation, and renders an interactive popup.	JavaScript, ServiceNow spModal API, Custom Portal Widget (metro_qr_widget)
5.	Database	Custom relational table storing station nodes, unique station abbreviations, and operational availability states.	ServiceNow Relational Database Schema (u_metro_station_details)
6.	Cloud Database	Cloud transactional storage tracking metro order fulfillment, passenger counts, generated sys_ids, and request items.	ServiceNow Cloud Backend Database (sc_request, sc_req_item)
7.	File Storage	Ephemeral storage and caching of the generated 250x250 scannable 2D barcode passes and portal widget assets.	ServiceNow Attachment Repository & Browser Cache / Blob Storage
8.	External API-1	Public REST endpoint for dynamic 2D barcode generation using encoded ticket parameters and unique sys_id tokens.	External QR Server API (https://api.qrserver.com/v1/create-qr-code/)
9.	External API-2	Identity verification interface for passenger validation and digital concession verification.	Government Identity Verification API / National ID Concession Interface
10.	Machine Learning Model	Predictive route and crowd density model forecasting peak station usage and optimal train dispatch schedules.	ServiceNow Predictive Intelligence / Transit Demand Forecasting Model

11.	Infrastructure (Server / Cloud)	Enterprise multi-instance cloud environment running the ServiceNow platform with automated scaling and daily backups.	Cloud Infrastructure: ServiceNow Advanced High Availability (AHA) Cloud Architecture
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Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Responsive client-side MVC framework and grid styling libraries powering the Service Portal and widget templates.	AngularJS 1.5.x, Bootstrap 3.3.x, FontAwesome Icons
2.	Security Implementations	Role-based Access Control Lists (ACLs) restricting write access to metro_admin, TLS 1.3 encryption in transit, and sys_id tokenization.	ServiceNow Contextual Security (ACLs), TLS 1.3 HTTPS, SHA-256 Payload Tokenization
3.	Scalable Architecture	Tiered multi-instance cloud model separating user interface, application script logic, and database clusters.	3-Tier Enterprise Cloud Architecture (Presentation, Logic Engine, Relational DB)
4.	Availability	Continuous system operation across redundant geographical cloud availability zones ensuring uninterrupted commuter access.	ServiceNow High Availability Pair (Dual Data Center Redundancy & Automated Failover)
5.	Performance	Sub-200ms client-side script execution for fare calculation, edge-cached UI styling, and asynchronous image retrieval.	Client-side Caching, Asynchronous REST API Calls, Optimized Indexed Database Queries